

Cybersecurity

PHISHING EMAIL DETECTION & AWARENESS REPORT (Task 2)

Date: May 1, 2026

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For: Future Interns

Executive Summary :

This report analyzes two real phishing emails collected from public security repositories. Both emails were examined using manual header analysis and MXToolbox to identify authentication failures and phishing indicators.

EMAIL #1 - 419 Advance Fee Fraud (Joseph Camarah Vieira)

This email promises \$60 million USD in exchange for assistance transferring funds from Guinea Bissau. Key red flags include: a Reply-To address (carr444@yahoo.com) that differs from the From address (vieria@aol.com), poor grammar ("he was short dead"), generic greeting ("Dear Sir/Madam"), and undisclosed recipients. MXToolbox revealed SPF neutral, no DKIM signature, and a spam score of 73 - extremely high.

EMAIL #2 - Brand Impersonation Scam (Fake Lowe's / DeWalt Giveaway)

This email pretends to be from Lowe's offering a free DeWalt Power Station. The sender domain (eagleacct.com) has no connection to Lowe's (lowes.com). MXToolbox found no DMARC record, no DKIM signature, and Microsoft's spam filter assigned a BCL:9 score - the highest spam confidence level.

CONCLUSION

Both emails are confirmed phishing attempts. Email #1 follows the classic "419 scam" pattern designed to steal money through advance fees. Email #2 follows the "free gift" pattern designed to steal personal information through fake surveys.

KEY STATISTICS

- Total emails analyzed: 2
- Confirmed phishing: 2 (100%)
- Authentication failures found in both emails
- Both emails received high spam scores from email providers

All evidence, including header extracts, email body screenshots, and MXToolbox results, is provided in the appendix.

INTRODUCTION :

Phishing attacks remain one of the most common cyber threats facing businesses today. Attackers use fake emails to trick users into sharing passwords, clicking malicious links, or sending money.

This report analyzes two real phishing email samples. Each email is examined for red flags including sender domain, authentication results, and content indicators. Technical analysis was performed using MXToolbox. Prevention guidelines are provided to help users identify future phishing attempts.

The emails analyzed are:

- **Email #1:** 419 Advance Fee Fraud (Joseph Camarah Vieira)
- **Email #2:** Brand Impersonation Scam (Fake Lowe's / DeWalt Giveaway)

Analysis :

– Email #1 (Joseph Scam)

Email Sample and Headers

```
Dear Sir/Madam,

my name is Joseph Camarah Vieira, i am from Guinea Bissau, my late father was the former minister of mines in my country Guinea Bissau, he was short dead by the rebels in my country, before his death he deposited $60 million Dollars with Global Trust Security company Accra Ghana, i want you to help me receive this money in your country for investment in your country. i will give you 30% of the total sum when the funds arrive your country.

Regards,
Mr Joseph Camarah Vieira
00233 244 617 863
my email:carrr444@yahoo.com|
```

```
Received: from BL2PR00711HT001.namprd07.prod.outlook.com (10.255.104.164) by
B12PR00711HT003.namprd07.prod.outlook.com (10.255.88.166) with Microsoft SMTP
Server (TLS) id 14.16.257.4; Thu, 17 Jan 2013 23:35:35 +0000
Received: from BL2PR00711HT002.namprd07.prod.outlook.com (10.255.104.165) by
BL2PR00711HT001.namprd07.prod.outlook.com (10.255.104.164) with Microsoft
SMTP Server (TLS) id 14.16.257.4; Thu, 17 Jan 2013 23:35:34 +0000
Received: from mail240-tx2-R.bigfish.com (65.55.88.116) by
BL2PR00711HT002.namprd07.prod.outlook.com (10.255.104.165) with Microsoft
SMTP Server (TLS) id 14.16.257.4; Thu, 17 Jan 2013 23:35:34 +0000
Received: from mail240-tx2 (localhost [127.0.0.1]) by mail240-tx2-R.bigfish.com (Postfix) with ESMTP id A050032025F for <jerryyp@mail.unomaha.edu>; Thu, 17 Jan 2013 23:35:33 +0000 (UTC)
X-Forefront-Antispam-Report: CIP=59.125.100.113;XIP=(null);UIP=(null);IPV=MLI;bf-shako.com.tw;RD=59-125-100-113;HINET-IP.hinet.net;EPVD=MLI
X-Bigfish: ps73(z27f52hd926zh1ee6h100h1cesh120zh1e76h1d1ah1d2ah158h18275bh12e12a8h66esh839h940h1ad2h1177h128sh12a5h12a9h12bdh137ah139eh13b6h13eah144h1537h162dh1631h1758h177fh184fh1898h300k503k9531wa7jk)
X-FOSE-spam: This message appears to be spam.
X-SpamScore: 73
Received-SPF: neutral (mail240-tx2: 59.125.100.113 is neither permitted nor denied by domain of aol.com) client-ip=59.125.100.113; envelope-from=vieria@aol.com; helo=bf-shako.com.tw; shako.com.tw ;
Received: from mail240-tx2 (localhost.localdomain [127.0.0.1]) by mail240-tx2
(MessageSwitch) id 1359465731454940_30539; Thu, 17 Jan 2013 23:35:31 +0000
(UTC)
Received: from TX2EWSW9007.bigfish.com (unknown [10.9.14.242]) by mail240-tx2.bigfish.com (Postfix) with ESMTP id 675424200E7 for <jerryyp@mail.unomaha.edu>; Thu, 17 Jan 2013 23:35:31 +0000 (UTC)
Received: from bf-shako.com.tw (59.125.100.113) by TX2EWSW9007.bigfish.com
(10.9.99.107) with Microsoft SMTP Server (TLS) id 14.1.225.23; Thu, 17 Jan
2013 23:35:28 +0000
Received: from mail.shako.com.tw (59.125.100.112;HINET-IP.hinet.net
[59.125.100.112]) by bf-shako.com.tw (6.14.3/8.14.3) with ESMTP id
r0HwXGm13928; Fri, 18 Jan 2013 07:34:12 +0800
X-Authentication-Warning: bf-shako.com.tw: Host 59-125-100-112.HINET-IP.hinet.net [59.125.100.112] claimed to be mail.shako.com.tw
Authenticated-By: nobody
X-SpamFilter-By: BOX Solutions Spamtrap 3.5 with qid r0HwXZ51028539, This message is passed by code: ctdos35128
Received: from User (85-250-54-29.bb.netvision.net.il[85.250.54.29])
(authenticated bits=0)
by mail.shako.com.tw (6.14.3/8.14.3) with ESMTP
id r0HwXZ51028539; Fri, 18 Jan 2013 07:33:38 +0800
X-Box-Message-Id: r0HwXZ51028539
Message-Id: C01301117223.r0HwXZ51028539@mail.shako.com.tw
X-Authentication-Warning: mail.shako.com.tw: Host 85-250-54-29.bb.netvision.net.il[85.250.54.29] claimed to be User
Reply-To: <carrr444@yahoo.com>
From: JOSEPH CAMARAH VIEIRA <vieria@aol.com>
Subject: [Spam-Mail] Dear Sir/Madam. (This message should be blocked: ctdos35128)
Date: Fri, 18 Jan 2013 01:46:07 +0200
Content-Type: text/plain; charset="Windows-1251"
Content-Transfer-Encoding: 7bit
X-Mailer: Microsoft Outlook Express 6.00.2600.0000
X-HMIMEInfo: Produced by Microsoft HIMEOLE V6.00.2600.0000
To: Undisclosed recipients;;
Return-Path: vieria@aol.com
X-MS-Exchange-Organization-SCL: 7
X-MS-Exchange-Organization-AVStamp-Mailbox: MSFTFF110;0 0 0
X-MS-Exchange-Organization-AuthSource: BL2PR00711HT002.namprd07.prod.outlook.com
X-MS-Exchange-Organization-AuthAs: Anonymous
MIME-Version: 1.0
```


Step	Delay	From	To	With	Time (UTC)	Blacklist
1	*	User 85.250.54.29				
2	*	mail.shako.com.bv 59.125.100.112	bf.shako.com.bv	ESMTP	1/17/2013 11:34:12 PM	✓
3	1 minutes	bf.shako.com.bv 59.125.100.113	TX2EHSMH8007.bigfish.com 10.9.99.107	Microsoft SMTP Server (TLS)	1/17/2013 11:35:28 PM	✓
4	3 seconds	TX2EHSMH8007.bigfish.com 10.9.14.242	mail240-b2.bigfish.com	ESMTP	1/17/2013 11:35:31 PM	✓
5	0 seconds	mail240-b2 127.0.0.1	mail240-b2		1/17/2013 11:35:31 PM	✓
6	2 seconds	mail240-b2 127.0.0.1	mail240-b2-R.bigfish.com	ESMTP	1/17/2013 11:35:33 PM	✓
7	1 Second	mail240-b2-R.bigfish.com 65.55.88.116	BL2PRD0711HT002.namprd07.prod.outlook.com 10.255.104.165	Microsoft SMTP Server (TLS)	1/17/2013 11:35:34 PM	✓
8	0 seconds	BL2PRD0711HT002.namprd07.prod.outlook.com 10.255.104.165	BL2PRD0711HT001.namprd07.prod.outlook.com 10.255.104.164	Microsoft SMTP Server (TLS)	1/17/2013 11:35:34 PM	✓
9	1 Second	BL2PRD0711HT001.namprd07.prod.outlook.com 10.255.104.164	BY2PRD0711HT003.namprd07.prod.outlook.com 10.255.88.166	Microsoft SMTP Server (TLS)	1/17/2013 11:35:35 PM	✓

Gmail & Yahoo are now requiring DMARC - Get yours setup with Delivery Center

SPF and DKIM Information

dmARC:aol.com Show Solve Email Delivery Problems <pre>v=DMARC1; p=reject; pct=100; rua=mailto:0@ua.agari.com; ruf=mailto:0@uf.agari.com;</pre>
spf:aol.com:59.125.100.113 Show Solve Email Delivery Problems <pre>v=spf1 redirect=_spf.mail.yahoo.com</pre>
Dkim Signature Error: No DKIM-Signature header found - more info
Dkim Signature Error: There must be at least one aligned DKIM-Signature for the message to be considered aligned. - more info

Test	Result	Explanation
SPF	Neutral	Domain (aol.com) neither approved nor denied the sender. Legitimate AOL emails would have SPF = Pass
DKIM	None	No digital signature found. AOL signs their legitimate emails
DMARC	None	No domain policy. AOL has DMARC policies for their real emails
Spam Score	73/100	Extremely high. Most legitimate emails score below 10
Spam Filter	" This message appears to be spam "	Microsoft's filter explicitly identified it as spam
Originating IP	85.250.54.29	Located in Israel - AOL is a US company

Attack Pattern Analysis:

This is a classic 419 Advance Fee Fraud (Nigerian Prince scam).

How it works:

- Attacker claims access to large sum of money (\$60 million)
- Offers victim percentage (30% = \$18 million)

- Once victim replies, attacker asks for "processing fees" or "legal fees"
- Victim pays fees, attacker disappears
- The \$60 million never existed

What the attacker wants: Money transfer fees (usually 500–5,000)

– EMAIL #2 (Lowe's/DeWalt - Brand Impersonation Scam)

Email Sample and Headers

```
Subject: Pending - Order [#21499-01]

Lowe's

Congratulations!

We would like to offer you a unique opportunity to receive a Free DeWalt Power Station!

To claim it, simply take this short survey about your experience with us.

CONTINUE FOR FREE

click here if you want to Unsubscribe.
```

```
From: Lowe's <qydnqycysp@eagleacct.com>
Subject: Pending - Order [#21499-01]
To: phishing@pot
Authentication-Results: spf=pass (sender IP is 52.100.155.227) smtp.mailfrom=eagleacct.com; dkim=fail (signature did not verify) header.d=txbo.onmicrosoft.com; dmarc=bestguesspass
action=none header.from=eagleacct.com
Received-SPF: Pass (protection.outlook.com: domain of eagleacct.com designates 52.100.155.227 as permitted sender)
X-MS-Exchange-Organization-SCL: 5
X-Microsoft-Antispam: BCL:9
```

Detailed Phishing Indicators			
#	Indicator	Evidence	Why is this a Red Flag
1	Fake sender domain	eagleacct.com instead of lowes.com	Lowe's official domain is lowes.com, not eagleacct.com
2	Spoofed display name	"Lowe's" uses special characters	The letters look normal but are actually different characters to bypass filters
3	Order number pretext	Subject: "Pending - Order [#21499-01]"	Creates urgency about a fake order
4	Too good to be true	"Free DeWalt Power Station"	DeWalt power stations cost 500–1000+
5	Generic offer	No name, no order details	Real Lowe's orders include specific item details

6	Fake Links	Links to digitaloceanspaces.com	Lowe's uses lowes.com, not file hosting services
7	Tracking pixel	asnolux.com/track/	Used to track if you opened the email

MXToolbox Technical Analysis:

Delivery Information

- DMARC Compliant (No DMARC Record Found)
 - SPF Alignment
 - SPF Authenticated
 - DKIM Alignment
 - DKIM Authenticated

Relay Information

Received Delay:

0 seconds

SPF and DKIM Information

dmARC:eagleacct.com

Hide

Solve Email Delivery Problems

Test	Result	
DMARC Record Published	No DMARC Record found	More Info

Reported by ns1071.ui-dns.org on 5/1/2026 at 2:05:58 PM (UTC 0), just for you

spf:eagleacct.com:52.100.155.227

Show

```
v=spf1 include:_spf-us.ionos.com include:spf.protection.outlook.com ~all
```

Dkim Signature Error:
No DKIM-Signature header found - [more info](#)

Dkim Signature Error:
There must be at least one aligned DKIM-Signature for the message to be considered aligned. - [more info](#)

Headers Found

Header Name	Header Value
From	Lowe's <qydnqycysp@eagleacct.com>
Subject	Pending - Order [#21499-01]
To	phishing@pot
Authentication-Results	spf=pass (sender IP is 52.100.155.227) smtp.mailfrom=eagleacct.com; dkim=fail (signature did not verify) header=d-bbo.onmicrosoft.com; dmarc=bestguesspass action=none headerfrom=eagleacct.com
Received-SPF	Pass (protection.outlook.com: domain of eagleacct.com designates 52.100.155.227 as permitted sender)
X-MS-Exchange-Organization-SCL	5
X-Microsoft-Antispam	BCL9

Test	Result	Explanation
SPF	Pass	Passed for eagleacct.com, NOT for lowes.com

DKIM	No signature found	Email not digitally signed. Lowe's signs their real emails
DMARC	No record found	Eagleacct.com has no DMARC policy. Legitimate retail domains have DMARC
BCL	9/9	Microsoft Bulk Complaint Level - maximum spam confidence
SCL	5/9	Microsoft Spam Confidence Level - moderately high

Attack Pattern Analysis:

This is a Brand Impersonation / Free Gift Scam.

How it works:

- Attacker impersonates a trusted brand (Lowe's)
- Offers a valuable free gift (DeWalt Power Station)
- Victim clicks link and enters personal information
- Information is stolen for identity theft or sold to other scammers

What the attacker wants: Personal information (name, address, phone, email, sometimes credit card for "shipping fee")

PREVENTION GUIDELINES :

DO'S

1. Check the sender's email address carefully

Attackers use fake domains like eagleacct.com instead of lowes.com. Always verify the domain matches the company.

2. Hover over links before clicking

The visible text may say "Lowe's" but the actual link goes to a fake website. Hover to see the real destination.

3. Verify offers directly on company websites

If Lowe's offers a free gift, go to lowes.com directly - don't click email links.

4. Look for poor grammar and spelling

"he was short dead" - legitimate companies proofread their emails.

5. Check for personalized greetings

"Dear Sir/Madam" or "Dear Customer" is suspicious. Real companies use your name.

6. Report suspicious emails to IT or security team

Reporting helps protect other employees from the same scam.

7. Enable two-factor authentication (2FA)

Even if attackers steal your password, 2FA blocks them from accessing your account.

DON'TS

1. Don't click links in unsolicited emails

Links lead to fake websites that steal your information.

2. Don't reply to suspicious emails

This confirms your email address is active, leading to more spam.

3. Don't trust too-good-to-be-true offers

60 million or free 1,000 products are always scams.

4. Don't ignore mismatched Reply-To addresses

If From and Reply-To are different, the attacker controls the reply address.

5. Don't enter personal information from email links

Fake websites look real but steal your data.

6. Don't forward suspicious emails to coworkers

You might spread the scam. Report to IT instead.

REMEMBER THIS RULE :

"If an email creates urgency, offers something too good to be true, or asks you to click a link - it's probably a phishing scam."

Legitimate companies will NEVER ask for your password via email or promise millions of dollars to a stranger.